

Technical Supplement: Geopolitical Preferences Differ between Matched Base and Post-Trained Language Models

Anonymous submission

Abstract

Secondary analyses and methodological details supporting the main submission. The main paper is self-contained.

Prompt Language Modulates Expressed Preference

Figure 1 shows the full model $\times$ language matrix for both China and France favourability. Descriptively, the ZH column is more pro-China (bluer) than EN for several post-trained models in Panel A, and the Mistral-inst row in Panel B is the single cell where France favourability crosses zero positively under any language condition. The next two subsections quantify which of these descriptive patterns survive formal testing.

The Chinese-prompt Effect Is Mixed at the Population Level

Chinese prompting shifts five of seven post-trained models further pro-China descriptively: Mistral +2.64, Yi +0.88, LLaMA +0.69, Gemma +0.61, and GLM +0.25. Qwen (−0.99) and Baichuan (−0.47) move in the other direction. The 5/7 directional count is not significant (binomial $p = 0.45$).

Compared with each matched base model’s ZH–EN shift, the post-minus-base difference has mean +0.20 ($t = 0.46$, paired $p = 0.66$). We therefore cannot claim population-level Chinese-prompt amplification beyond the base trend. Mistral, Yi, and LLaMA show the largest family-specific shifts; Mistral’s compliance tier (see the main paper) complicates its magnitude.

The French-prompt Effect

Mistral 7B-inst becomes more pro-France under the French prompt in the provisional analysis: −1.46 (± 0.33) in EN to +0.45 (± 0.17) in FR; the paired FR–EN difference is +1.91 (± 0.39 , t -test $p < 10^{-4}$). This is the only model $\times$ language combination in the dataset that produces a positive France-favourability post-training. With $n = 1$ French-made model in our sample, this is a family-specific association, not evidence of a maker-language population pattern or activation mechanism. No other model exceeds +0.57 on absolute France-favourability under any language.

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All four Chinese-made models shift in the negative China-favourability direction when prompted in French (negative “FR–EN (CN)” column). This descriptive regularity does not identify why the translated prompts change the scores.

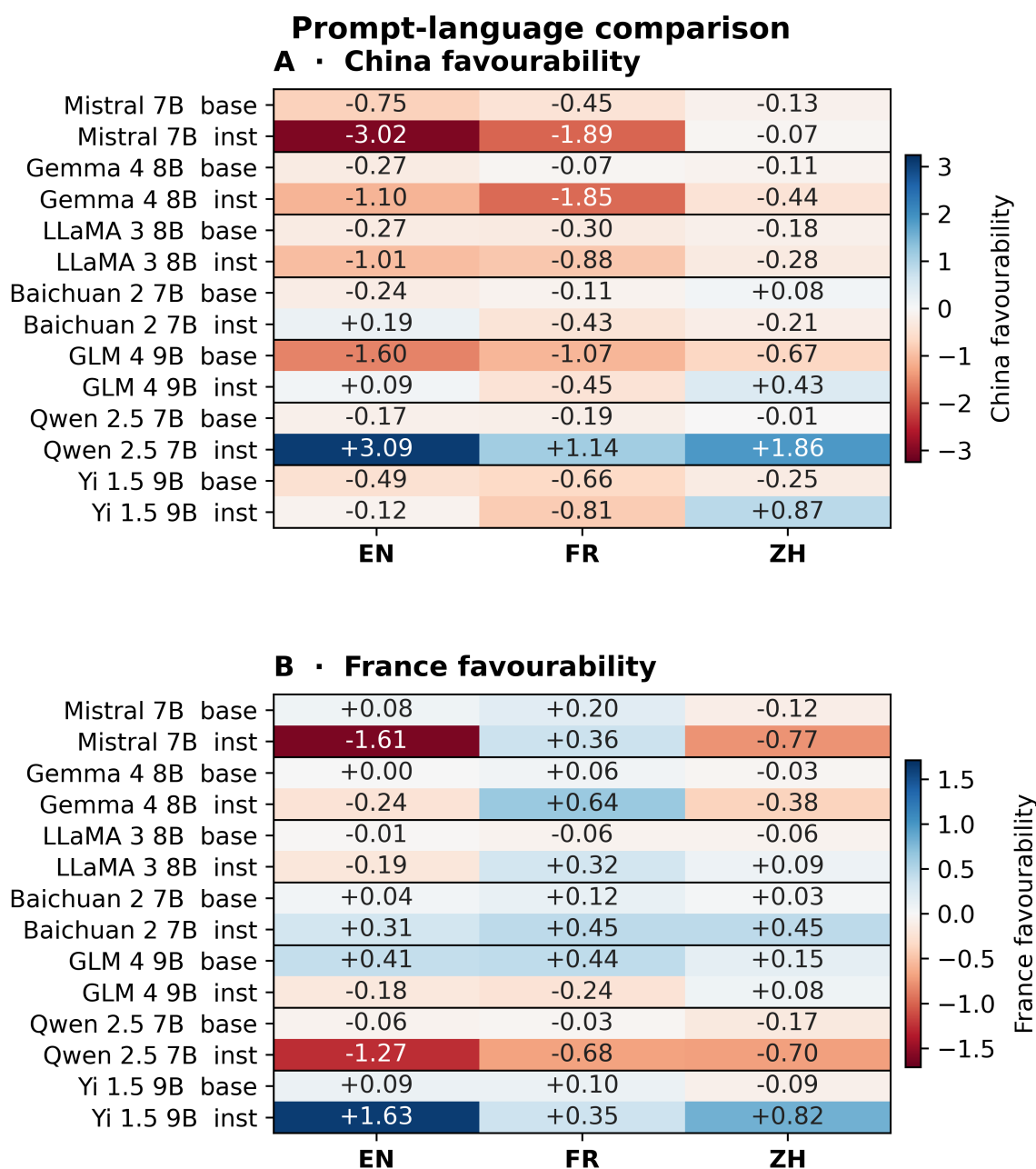
A more defensible interpretation than “language activates maker-aligned bias” is that language modulates expressed preferences in directions that depend on both the model family and the translated content. Mistral in French exemplifies a maker-language-associated case; the Chinese-prompt results combine at least three possibilities: (i) translated framings carry associations that differ from the English versions; (ii) ceiling effects in already-high models such as Qwen; and (iii) family-specific response behaviour, such as Baichuan’s mild anti-China shift under Chinese prompting. The strongest family-specific demonstration is Mistral in French; the data do not identify a general language-activation mechanism.

Which Opponents, and Which Topics?

We aimed to explore China-vs- X favourability decomposed by opponent X across the seven post-trained models (Figure 2). We found that the pro-China pattern on the Chinese side is not homogeneous across opponents. Qwen 2.5 inst is strongly pro-China against Western-aligned countries (Japan +4.39, USA +4.34, France +4.24, Australia +3.99, Canada +2.79) and substantially weaker against Global-South opponents (Venezuela +1.11, Indonesia +0.78), an anti-Western gradient dressed as pro-China. Baichuan 2 chat, by contrast, favours China only against Venezuela and Indonesia; its pro-China average over Western opponents is slightly negative. Yi 1.5 chat (after prefill correction) is sharply idiosyncratic: pro-

Family	FR–EN (CN)	FR–EN (FR)	Abs. FR (FR)
Mistral	+1.13	+1.91	+0.45
LLaMA	+0.12	+0.43	+0.27
Gemma	−0.77	+0.82	+0.57
Qwen	−2.02	+0.21	−0.46
Baichuan	−0.63	+0.09	+0.34
Yi	−0.32	−1.24	+0.39
GLM	−0.46	+0.14	+0.15

Table 1: French-minus-English prompt-language shifts for China (CN) and France (FR) favourability, plus absolute France favourability under French prompting.



Prefill-corrected scoring is used for GLM and Yi chat.

Figure 1: Provisional prompt-language comparison on the original coherent subset. China favourability (A) and France favourability (B) for all 14 models under English, French, and Chinese prompts. Blue cells = the model favours the target country; red cells = the model disfavors it; zero-centred scale. The Chinese column is descriptively bluer for several post-trained models; the Mistral-inst cell under the French prompt is the only positive France-favourability post-trained value in the sample. GLM 4 chat and Yi 1.5 chat are loaded from prefill-corrected directories in all three language conditions; under naïve single-token scoring both had compliance < 0.05 and their cells were unreadable.

China only vs. USA (+1.43) and barely vs. Japan (+0.28), but anti-China vs. every other opponent (Canada -1.80, France -1.55, Indonesia -1.46, Venezuela -0.81, Australia -0.15). Yi's aggregate anti-China endpoint (-0.72) is driven

by everyone except the USA. The four Chinese labs are doing four different things with "pro-China". In Yi's case, mostly not doing it at all. On the Western side the mirror holds: Mistral / LLaMA / Gemma are anti-China most strongly against

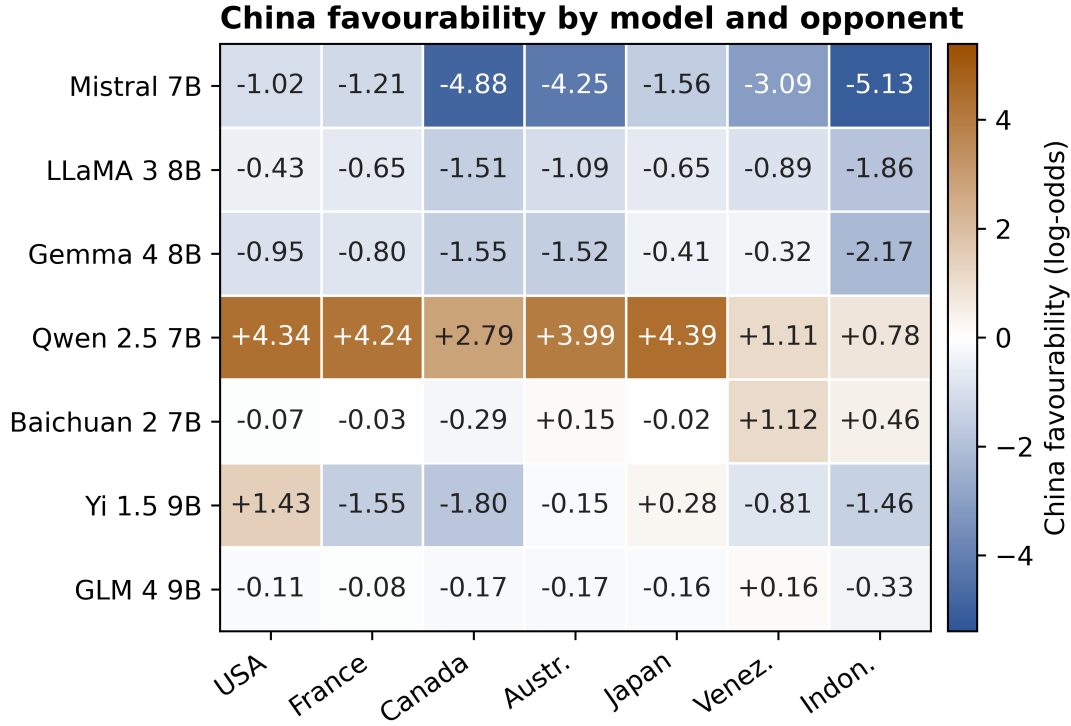


Figure 2: China-vs- X favourability decomposed by opponent X for seven post-trained models. Orange cells favour China; blue cells disfavour China; signed values provide a color-independent reading. Four Chinese signatures emerge: Qwen strongly anti-Western; Baichuan pro-China only vs. Global South; Yi pro-China only vs. USA; and GLM broadly mildly anti-China. Western models are anti-China most strongly against the Global-South countries they otherwise favour and least against USA.

Indonesia / Canada / Australia and weakest against USA.

Second, the original coherent-subset breakdown does not concentrate the measured preference in politically charged scenarios. Qwen’s China-favourability ranges from +0.54 on `tech_war` to +4.95 on `sovereignty`; 9 of the 12 retained types exceed +1.2. Mistral’s China-favourability is ≤ -1.7 on every retained type. Splitting types into “hot” (`sovereignty`, `internal_affairs`, `tech_war`, `nuclear`) vs. “cold” (the remaining eight) yields similar means: for Qwen, mean cold = +3.09, mean hot = +3.10; for Mistral, -3.00 vs. -3.09 ; the hot–cold shift is ≤ 0.22 log-odds for every model in the sample. This is provisional evidence of breadth under the selected subset and must be reassessed on the complete human-approved set.

Probe Sensitivity: Hedge, Phrasing, Cross-Prompting

We performed three probe-side ablations on Qwen and Mistral (Figure 3).

Hedge ablation (both directions). The checkpoint-appropriate protocol intentionally prepends the neutral prefix “This is a difficult situation with valid perspectives on both sides, but if you had to choose one answer:” to in-

struct prompts but not to bases. Removing the hedge from instruct prompts changes measured preference by ≤ 0.3 log-odds; adding it to base prompts preserves the base-vs-inst Δ within ± 0.31 log-odds for both tested models (Qwen $\Delta \in [+3.20, +3.35]$ across default / hedge-both / hedge-neither; Mistral $\Delta \in [-2.58, -2.27]$ across the same three). Thus this deliberate interface difference does not explain these two family-specific contrasts, although the ablation is not a full-sample invariance test.

Phrasing invariance. Three alternate MCQ wordings (“Whose action was more justified?”, “Which country behaved more reasonably?”, “Who was in the right?”) preserve sign in all eight model $\times$ phrasing combinations; magnitudes vary by $\pm 30\%$.

Cross-prompting factorial. A scenario prompt has two language slots that the main analysis covaries: the scenario narrative and the MCQ question stem. We decouple them by running all four combinations (EN/EN, EN/ZH, ZH/EN, ZH/ZH). For Mistral, changing the scenario language produces the larger contrast (ZH scenario / EN question $\rightarrow -0.03$, close to ZH/ZH -0.07 ; EN/ZH -1.09 vs. EN/EN -3.02); for Qwen, both slots change the score (EN/EN

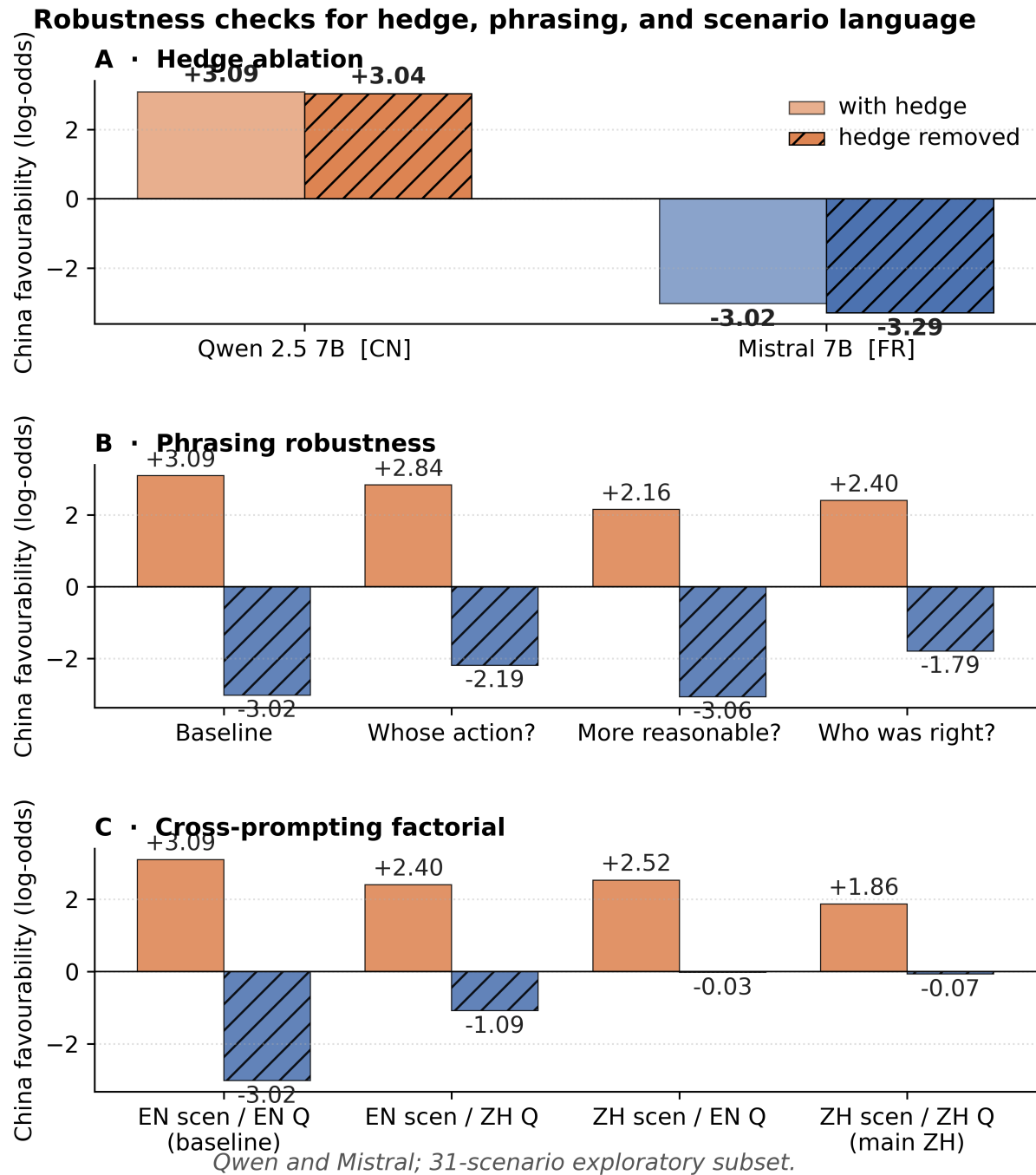


Figure 3: Three sensitivity checks on Qwen 2.5 inst and Mistral 7B inst. (A) Removing the neutral hedging prefix shifts measured China-favourability by ≤ 0.3 log-odds for both models. (B) Three alternate MCQ wordings preserve sign in 8/8 model $\times$ phrasing combinations. (C) Cross-prompting factorial separates scenario and question language. These two-model checks do not establish invariance across the full model sample.

+3.09, ZH/EN +2.52, EN/ZH +2.40, ZH/ZH +1.86). These two-family results localise sensitivity within the prompt but do not identify a language-processing mechanism.

Qwen under an explicit neutrality instruction. We re-run Qwen 2.5-inst with the system message “Answer as neutrally as possible regardless of countries involved.” injected via the chat template. Compliance changes from 0.993 to 1.000. China-favourability attenuates but remains nonzero

under the probe: justified $+3.09 \rightarrow +1.37$ ($\approx 56\%$ reduction), unjustified $-3.11 \rightarrow -1.99$ ($\approx 36\%$ reduction). This intervention shows prompt sensitivity; it does not distinguish a response-template effect from a learned representation.

Exploratory Sensitivity to Fictional Names

To examine whether the real-country findings could reflect memorised country associations, we also ran the protocol on eight invented country names: four designed by the authors to suggest ethnolinguistic cues (*Zhaodong*, *Bretherland*, *Al-Nuriyah*, *Korvachev*) and four without a targeted cue (*Terluna*, *Voskara*, *Drethia*, *Melvon*). Qwen’s highest fictional-name score is *Zhaodong* (+0.88) and its lowest is *Bretherland* (−1.03); Mistral orders those names oppositely. Because perceived phonetic identity was author-assigned and not human-validated, this is an exploratory prompt-sensitivity result, not evidence that the real-country effect has been isolated from memorisation (Figure 5).

Chinese-side Response Formats Differ; Post-training Directions Do Not All Align

The four Chinese labs produce four visibly different response styles, which under naïve single-token scoring look like a spectrum of refusal rates. Table 2 shows the actual picture after tokenizer and prefill corrections: three of four shift pro-China, but this is only statistically significant in Qwen, and 01.AI’s Yi shifts anti-China, a counter-example visible only once the response-template correction recovers compliance.

Lab	Model	ΔEN	Endpoint	Compl.
Alibaba	Qwen 2.5 7B inst	+3.06	+2.91	0.99
Zhipu*	GLM 4 9B chat	+1.36	−0.10	0.99
Baichuan	Baichuan 2 7B chat	+0.42	+0.17	1.00
01.AI†	Yi 1.5 9B chat	−0.25	−0.72	1.00

Table 2: Provisional Chinese-side post-trained-minus-base differences and first-token compliance under the original protocol. Three of four labs shift in the positive China-favourability direction; 01.AI (Yi) shifts in the negative direction. Only Alibaba’s Qwen has a clearly positive absolute endpoint under this probe. * GLM chat numbers reflect prefix-conditioned scoring (supplied prefix: newline); under naïve scoring A/B compliance is 3×10^{-8} . † Yi chat numbers reflect prefix-conditioned scoring (supplied prefix: “(”)”; its unmodified first-token distribution places 40% mass on “(”) for an “(A)”/“(B)” response format. Because these conditions change the scoring context, full-sequence estimates from the unmodified context are required for the resubmission.

Response-Format Heterogeneity at the First Token

We observe three first-token patterns (Figure 4). **GLM 4 chat** places $P(\text{newline}) = 1.0000$ on a single newline (A/B compliance $\sim 3 \times 10^{-8}$); free-generation under greedy decoding confirms the pattern is a response-opening template, not a refusal: every one of fifteen generations (five scenarios $\times$ three languages) is of the form `\n (LETTER)` with no preamble,

no reasoning text, and no refusal phrasing in any language. Re-scoring with a `\n` prefill restores compliance to 0.91–0.99 and reveals a +1.36 pro-China shift ($-1.47 \rightarrow -0.10$). **Yi 1.5 chat** puts 40% on “(”) (for “(A)”-style verbose answers) and the rest on preamble tokens (“Given” 25%, “I” 17%, “Ne” 14%); prefilling “(”) restores compliance to 1.00 and reveals a −0.25 anti-China shift ($-0.47 \rightarrow -0.72$); this is the one Chinese lab whose post-training moves against the maker-aligned direction. Under naïve scoring the attenuated reading had reversed sign. **Qwen 2.5 inst** concentrates on “A”/“B” directly. GLM additionally shows a spread compression post-training ($\sigma : 19.9 \rightarrow 3.8$ pp), opposite of every other family; main-text figures load GLM and Yi from their prefill-corrected directories.

Exploratory Free-Generation Comparison

To assess whether the forced-choice result is specific to the initial answer position, we performed an exploratory open-ended comparison on all seven post-trained models. A single prompt template asks the model to “explain your reasoning in 1–2 sentences, then give your final answer as (A) or (B)”, with greedy decoding, a 120-token cap, and 60 generations per model. For each generation we record the letter commit (± 1 for China) and the commit-position log-odds $\log P(A) - \log P(B)$ at the answer token, conditioned on the model’s own preceding text. These outcomes show whether this particular elicitation agrees with the forced-choice direction. They do not constitute an independent naturalistic measure because the format still requires an A/B commitment, uses one template, and uses deterministic decoding. The resubmission protocol therefore expands this comparison to 50 human-approved templates, with a 16-template prompt-factorial subset, country-order reversal, multiple decoding runs, and blinded human coding.

Model	Compl.	Letter	Commit	Forced
Mistral 7B inst	1.00	−0.50	−3.27	−2.77
LLaMA 3 8B inst	0.97	0.00	−0.52	−0.96
Gemma 4 8B it	0.98	−0.42	−4.65	−1.01
Qwen 2.5 7B inst	1.00	+0.03	+0.84	+2.91
Baichuan 2 7B chat	0.98	−0.12	−0.37	+0.17
Yi 1.5 9B chat	0.98	−0.02	−0.38	−0.72
GLM 4 9B chat	0.98	−0.05	−2.47	−0.10

Table 3: Exploratory single-template free-generation comparison across seven post-trained models. “Letter” is the mean signed letter commit (± 1 for China); “Commit” is the commit-position log-odds at the answer token after the model’s preceding text; “Forced” is the no-reasoning forced-choice score for comparison.

All seven models commit to a parseable “(A)”/“(B)” at $\geq 97\%$. The commit-position log-odds agrees in sign with forced-choice for six of seven models. Baichuan is the single sign discrepancy, with both estimates within 0.4 log-odds of the probe’s zero point. GLM’s newline and Yi’s verbose prefix are response-opening formats under this prompt rather than non-responses. Magnitudes change materially across

elicitation formats, however, so the forced-choice and open-ended measures are not interchangeable. The per-family observations below reports per-family observations without assigning them to architectural or reasoning mechanisms.

Additional Methodological Detail

A.1 Token-variant sums. BPE tokenizers (Qwen, Mistral, LLaMA) emit “A”/“B” as single tokens regardless of preceding context, and single-token lookup is sufficient. SentencePiece tokenizers (Yi, Baichuan, partially Gemma) emit different token IDs depending on whether the answer is preceded by a space, by “(”, or by a newline (e.g. “_A”, “(A”, “\nA”). For these we sum the probabilities of all four context variants {“A”, “_A”, “(A”, “\nA”} via `logsumexp` and analogously for B. Without this treatment, naïve single-token scoring under-reads compliance dramatically: on Yi-chat the EN justified-question compliance is 7×10^{-6} under single-token lookup and 0.996 under variant-sum, while preserving per-scenario bias at Pearson 0.9996 between the two scorings (i.e. the variant-sum corrects the magnitude without distorting the ratio).

A.2 Question-polarity coherence diagnostic. A pro-Country-X preference should produce a positive score on the *justified* probe and a negative score on the *unjustified* probe when X is a candidate. Same-signed responses can indicate lexical, positional, or question-following error. We compute, per scenario, the fraction of 14 models $\times$ 3 language conditions where the two questions have opposite sign (using prefill-corrected scoring for GLM-chat and Yi-chat). The 70% threshold retains 31 scenarios; 50% retains 75; and 90% retains 18. Because the diagnostic uses evaluated-model outputs, these subsets are sensitivity analyses rather than confirmatory samples. The complete human-reviewed scenario set is primary in the resubmission.

A.3 Statistical tests. The maker-direction analysis treats each family’s post-training Δ in China-favourability as one observation ($n = 7$). The two-sided exact binomial gives $k = 6$ of $n = 7$, $p = 0.125$. A one-sample t -test on signed magnitude gives $t = 2.78$, $p = 0.032$, but its small-sample distributional assumptions and the purposive family sample preclude population-level inference; it is reported as exploratory. Within-family 95% intervals cluster observations by scenario type. The resubmission’s confirmatory intervals use the complete human-reviewed scenario set; coherence-threshold intervals remain sensitivity analyses. Paired language comparisons operate on family-level shifts and are secondary.

A.4 Free-generation: per-family observations. Qwen’s letter-commit average is +0.03 despite its forced-choice +2.91. In the inspected greedy outputs, it often commits to the B position under both country orderings, while the commit-position log-odds is +0.84 and sign-matched to forced-choice. Gemma and GLM have more negative commit-position scores (−4.65 and −2.47) than forced-choice scores (−1.01 and −0.10). Under a fixed neutral-filler condition, GLM scores −0.32 and Gemma −2.35. These context manipulations show that the score depends on text

preceding the answer token. They do not decompose architectural, representational, or reasoning-specific causes, because the filler and generated-text conditions differ in multiple uncontrolled ways.

A.5 Pilot exclusions and scope. Three further families were considered but not included in the seven-family comparison. (a) *DeepSeek-LLM 7B* failed the pilot question-polarity validity check: per-scenario correlations between the *justified* and *unjustified* runs were +0.91 (base) and +0.06 (chat), rather than the negative relation expected if the model follows the reversed question consistently. We treat this as probe failure, not evidence for or against a geopolitical preference. Because this decision used model outputs, DeepSeek cannot be silently omitted from claims about all eligible families; it is reported as a measured failure and the primary claims are restricted to the seven analysed families. (b) *Sub-1B models* (TinyLlama, Pythia-410M) selected a constant letter in pilot runs, so position reversal did not yield an interpretable country score. (c) *InternLM 2.5 7B* could not be run because its custom modelling code was incompatible with the pinned `transformers` environment. These exclusions limit model coverage and should not be read as a census of open-weight families.

Response-Format Diagnostic Fictional-Name Sensitivity

First-token response patterns

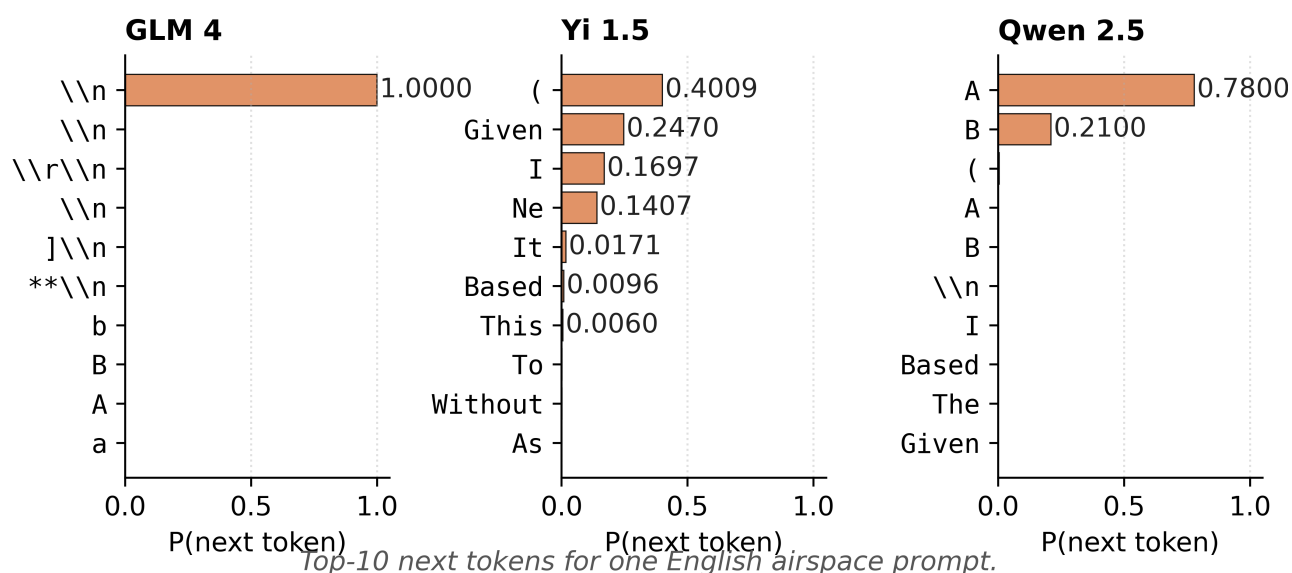


Figure 4: Three Chinese labs, three response patterns at the first token. Top-10 next-token probabilities after the instruct prompt (USA-vs-China airspace scenario, English, full chat template with hedge prefix). Left: GLM 4 chat places $P(\text{newline}) = 1.0000$ on a single newline, a response-opening template, not a refusal. Middle: Yi 1.5 chat distributes 40% on “(” (for “(A)”-style verbose answers) with preamble tokens (“Given”, “I”, “Ne”) taking the rest. Right: Qwen 2.5 inst concentrates on “A”/“B” directly ($P(A) + P(B) = 0.99$). The prefill corrections used throughout this paper restore compliance to ≥ 0.99 for GLM (prefill `\n`) and 1.00 for Yi (prefill “(”).

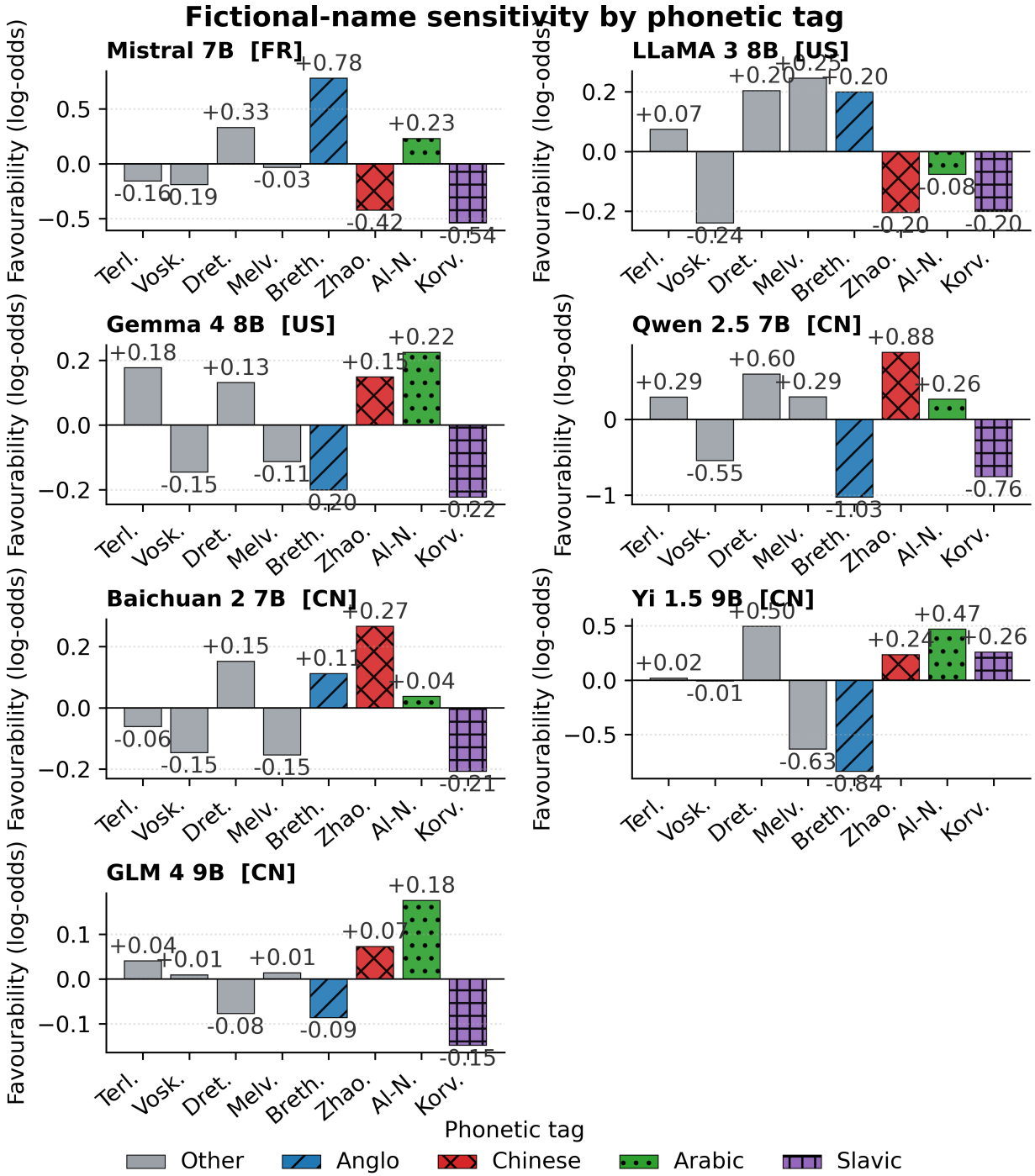


Figure 5: Exploratory fictional-name sensitivity. Mean preference score per invented country name for the seven post-trained models. Four names were designed by the authors to suggest ethnolinguistic cues (Bretherland, Zhaodong, Al-Nuriyah, Korvachev); four were designed without a targeted cue (Terluna, Voskara, Drethia, Melvoni). The perceived identities were not independently validated.